

The Constants of Nature

$E_{\rm h}:\text{K}=\frac{\text{kelvin}}{\text{K}}\left(\frac{T_{\rm p}}{m_{\rm p}}\right)\left(1-14\,C_{\rm U}^{-4}\,\mathscr{K}_{\Theta}\right)\boxtimes$	$m_{\rm de}=\frac{3s}{\mathscr{K}_3\,\mathscr{K}_4}\left(\frac{16}{4\pi}\right)\left(m_{\rm n}+m_{\rm +}\right)\left(1+\frac{Im(\omega_1)}{\mathscr{K}_2}\right)\left(\frac{L_2}{L_1}\right)\,\mathscr{K}_{\gamma}^{-3}\boxtimes$	$\mathcal{E}_{\rm p}=\frac{\text{joule}}{\text{kg}}\left(\frac{t_{\rm p}^2\,m_{\rm p}}{t_{\rm p}^2}\right)\left(1-\frac{1}{2}\,Re\left(i^{t^{''}}\right)\right)\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes$	$\frac{\mu_{\rm h}}{hc}=\frac{\mathscr{K}_1}{4\pi}\left(\frac{t_{\rm p}}{t_{\rm p}}\frac{q_{\rm p}}{m_{\rm p}}\right)\left(1+\sqrt{\frac{1}{D_{\rm D0}}}\right)\,\mathscr{K}_1\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes$	$c=\left(\frac{L_2}{T_{\rm p}}\right)\left(1+\sqrt{G_{\rm G1}}\right)\left(\mathscr{K}_1-\mathscr{K}_2\right)\boxtimes$	$F_{90}=\text{farad}\left(1+\frac{C_d}{6}\left(\mathscr{K}_1+\mathscr{K}_3+\mathscr{K}_4\right)\right)\boxtimes$	$K_j=\frac{\mathscr{K}_1}{\pi}\left(\frac{t_{\rm p}}{l_{\rm p}}\frac{q_{\rm p}}{m_{\rm p}}\right)\left(1-\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\left(\mathscr{K}_1+\mathscr{K}_2\right)^2\right)\boxtimes$	$g_{\rm de}=\frac{2\pi}{\text{Hz}}\left(1+\frac{16}{\rho_{\rm p}}\left(\mathscr{K}_1^2+\mathscr{K}_3^2+\mathscr{K}_4^2\right)\right)\boxtimes$
$\text{K}:\text{E}_{\rm h}=\frac{\text{kelvin}}{\mathscr{K}_1^4}\left(\frac{m_{\rm p}}{t_{\rm p}}\frac{m_{\rm p}}{m_{\rm e}}\right)\left(1+14\,C_{\rm U}^{-4}\,\mathscr{K}_{\Theta}\right)\boxtimes$	$m_{\rm u}=\frac{1}{\mathscr{K}_3\,\mathscr{K}_4}\sqrt{\frac{16}{4\pi}}\left(m_{\rm n}+m_{\rm +}\right)\left(1+\left(\frac{16}{2\pi}\right)\,\mathscr{K}_{\gamma}^{-3}\boxtimes\right)$	$R_{\rm K}=\frac{2\pi}{\mathscr{K}_1^2}\left(\frac{t_{\rm p}^2\,m_{\rm p}}{t_{\rm p}}\frac{m_{\rm p}}{q_{\rm p}^2}\right)\left(1+\frac{1}{2}\,Re\left(i^{t^{''}}\right)\right)\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes$	$\frac{\mu_{\rm N}}{hc}=\frac{\mathscr{K}_1}{4\pi}\left(\frac{t_{\rm p}}{t_{\rm p}}\frac{q_{\rm p}}{m_{\rm p}}\right)\left(1+\sqrt{\frac{1}{D_{\rm D0}}}\right)\,\mathscr{K}_1\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes$	$\frac{1}{\text{m}}\doteq\text{Hz}=\frac{1}{\text{meter}}\left(\frac{1}{t_{\rm p}}\right)\left(1+\sqrt{G_{\rm G1}}\right)\left(\mathscr{K}_1-\mathscr{K}_2\right)\boxtimes$	$H_{00}=\text{henry}\left(1-\frac{C_d}{6}\left(\mathscr{K}_1+\mathscr{K}_3+\mathscr{K}_4\right)\right)\boxtimes$	$\frac{e}{h}=\mathscr{K}_1\left(\frac{t_{\rm p}}{l_{\rm p}}\frac{q_{\rm p}}{m_{\rm p}}\right)\left(1-\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\left(\mathscr{K}_1+\mathscr{K}_2\right)^2\right)\boxtimes$	$\frac{\mu_{\rm de}}{8K}=\frac{2\pi}{8K}\left(1+\frac{16}{\rho_{\rm p}}\left(\mathscr{K}_1^2+\mathscr{K}_3^2+\mathscr{K}_4^2\right)\right)\boxtimes$
$\Delta_{\rm vcs}=\frac{E_1}{2\pi}\left(\frac{t_{\rm p}}{l_{\rm p}^2}\frac{m_{\rm p}}{m_{\rm p}}\right)\left(1-\sqrt{\zeta(2)}\right)\,\mathscr{K}_{\Theta}^2\boxtimes$	$m_{\rm s}=\frac{-2\pi\mathscr{K}_1}{\mathscr{K}_3\mathscr{K}_4}\left(\frac{\text{GeV}}{\sqrt{5}}\right)\left(\frac{t_{\rm p}^2}{l_{\rm p}^2}\right)\left(1+\frac{1}{6^{1/2}}\left(\frac{2\pi}{2s}\right)\,\mathscr{K}_{\gamma}^{-3}\boxtimes\right)$	$Z_0=4\pi\left(\frac{t_{\rm p}^2\,m_{\rm p}}{t_{\rm p}}\frac{m_{\rm p}}{q_{\rm p}}\right)\left(1+\frac{1}{2}\,Re\left(i^{t^{''}}\right)\right)\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes$	$A_{90}=\text{ampere}\left(1+\sqrt{\frac{9}{J_{0.1}}}\,\mathscr{K}_1\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes\right)$	$r_{\rm e}=\mathscr{K}_1^{-2}\left(\frac{t_{\rm p}}{m_{\rm p}}\right)\left(1-\sqrt{6\,C_{\rm R1}}\right)\left(\mathscr{K}_1-\mathscr{K}_2\right)\boxtimes$	$\Omega_{00}=\text{ohm}\left(1-\frac{C_d}{6}\left(\mathscr{K}_1+\mathscr{K}_3+\mathscr{K}_4\right)\right)\boxtimes$	$\Phi_0=\frac{\pi}{\mathscr{K}_1}\frac{t_{\rm p}^2}{2}\frac{m_{\rm p}}{q_{\rm p}}\left(1+\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\left(\mathscr{K}_1+\mathscr{K}_2\right)^2\right)\boxtimes$	$\frac{\mu_{\rm de}}{\mu_{\rm B}}=\frac{2\pi}{8K}\left(\frac{m_{\rm p}}{m_{\rm e}}\right)\left(1+\frac{16}{\rho_{\rm p}}\left(\mathscr{K}_1^2+\mathscr{K}_3^2+\mathscr{K}_4^2\right)\right)\boxtimes$
$\text{eV}:\text{Hz}=\frac{\text{eV}}{2\pi}\left(\frac{t_{\rm p}}{l_{\rm p}^2}\frac{m_{\rm p}}{m_{\rm p}}\right)\left(1-\sqrt{\zeta(2)}\right)\,\mathscr{K}_{\Theta}^2\boxtimes$	$m_{\rm he}=\frac{\sqrt{3}}{\mathscr{K}_1^2\,\mathscr{K}_2^2}\left(\frac{\text{GeV}}{2\pi}\right)\left(\frac{t_{\rm p}^2}{l_{\rm p}^2}\right)\left(1-\frac{3^{-1}}{\mathscr{K}_1\mathscr{K}_2}\left(\frac{s}{18}\right)\,\mathscr{K}_{\gamma}^{-3}\boxtimes\right)$	$Z_{\rm p}=\left(\frac{t_{\rm p}^2\,m_{\rm p}}{t_{\rm p}}\frac{m_{\rm p}}{q_{\rm p}}\right)\left(1+\frac{1}{2}\,Re\left(i^{t^{''}}\right)\right)\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes$	$C_{90}=\text{coulomb}\left(1+\sqrt{\frac{9}{J_{0.1}}}\,\mathscr{K}_1\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes\right)$	$a_0=\frac{1}{\mathscr{K}_1^2}\left(\frac{t_{\rm p}}{m_{\rm p}}\right)\left(1-\sqrt{6\,C_{\rm R1}}\right)\left(\mathscr{K}_1-\mathscr{K}_2\right)\boxtimes$	$ST_0=\frac{5}{2}+\log\left(\frac{1}{2\pi}\frac{1}{l_{\rm p}^2}\frac{K\,A_{\rm mass}}{T_{\rm p}\,m_{\rm p}}\right)^{3/2}\frac{K\,E_{\rm p}}{T_{\rm p}\,P_0}\left(1-\frac{1}{12}\left(\frac{s^2}{\mathscr{K}_1\mathscr{K}_2}\right)\left(\mathscr{K}_1-\mathscr{K}_3-\mathscr{K}_4\right)\right)\boxtimes$	$\frac{h}{e}=\frac{\text{coulomb}}{\mathscr{K}_1}\left(\frac{t_{\rm p}^2\,m_{\rm p}}{t_{\rm p}}\frac{m_{\rm p}}{q_{\rm p}}\right)\left(1+\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\left(\mathscr{K}_1+\mathscr{K}_2\right)^2\right)\boxtimes$	$b^{\prime}=\frac{v_{\rm peak}}{2\pi}\left(\frac{1}{t_{\rm p}\,T_{\rm p}}\right)\left(1+\frac{16}{16}\left(\frac{2\pi}{s}\right)^2\left(\mathscr{K}_1^2-\mathscr{K}_3^2-\mathscr{K}_4^2\right)\right)\boxtimes$
$\text{J}:\text{Hz}=\frac{\text{joule}}{2\pi}\left(\frac{t_{\rm p}}{l_{\rm p}^2}\frac{m_{\rm p}}{m_{\rm p}}\right)\left(1-\sqrt{\zeta(2)}\right)\,\mathscr{K}_{\Theta}^2\boxtimes$	$m_{\rm he}=\frac{\sqrt{3}}{\mathscr{K}_1^2\,\mathscr{K}_2^2}\left(\frac{\text{GeV}}{2\pi}\right)\left(\frac{t_{\rm p}^2}{l_{\rm p}^2}\right)\left(1-\frac{3^{-1}}{\mathscr{K}_1\mathscr{K}_2}\left(\frac{s}{18}\right)\,\mathscr{K}_{\gamma}^{-3}\boxtimes\right)$	$\text{Hz}:\frac{1}{\text{m}}=\frac{1}{\text{second}}\left(\frac{t_{\rm p}}{l_{\rm p}}\right)\left(1+\frac{1}{2}\,Im\left(i^{t^{''}}\right)\right)\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes$	$\frac{\mu_{\rm s}^{'}}{S}=\frac{2\,\mathscr{K}_1\mathscr{K}_2^2}{S}\left(1-18\left(6\pi\right)\mathscr{K}_1\mathscr{K}_3\mathscr{K}_4\boxtimes\right)$	$\sigma_{\rm e}=\sqrt{\frac{32}{18}}\,2\pi\mathscr{K}_1\left(\frac{t_{\rm p}}{l_{\rm p}}\frac{m_{\rm p}}{m_{\rm p}}\right)^2\left(1-2\sqrt{6\,C_{\rm R1}}\right)\left(\mathscr{K}_1-\mathscr{K}_2\right)\boxtimes$	$ST_1=\frac{5}{2}+\log\left(\frac{1}{2\pi}\frac{1}{l_{\rm p}^2}\frac{K\,A_{\rm mass}}{T_{\rm p}\,m_{\rm p}}\right)^{3/2}\frac{K\,E_{\rm p}}{T_{\rm p}\,P_1}\left(1-\frac{1}{12}\left(\frac{s^2}{\mathscr{K}_1\mathscr{K}_2}\right)\left(\mathscr{K}_1-\mathscr{K}_3-\mathscr{K}_4\right)\right)\boxtimes$	$\frac{h}{e}=2\pi\frac{\text{coulomb}}{\mathscr{K}_1}\left(\frac{t_{\rm p}^2\,m_{\rm p}}{t_{\rm p}}\frac{m_{\rm p}}{q_{\rm p}}\right)\left(1+\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\left(\mathscr{K}_1+\mathscr{K}_2\right)^2\right)\boxtimes$	$b^{\prime}=\frac{v_{\rm peak}}{2\pi}\left(\frac{1}{t_{\rm p}\,T_{\rm p}}\right)\left(1+\frac{16}{16}\left(\frac{2\pi}{s}\right)^2\left(\mathscr{K}_1^2-\mathscr{K}_3^$

$t_p = 5.3912583683213 \times 10^{-19}$ s	Planck time
$l_p = 1.61625918175645 \times 10^{-35}$ m	Planck length
$q_p = 1.87554596713962 \times 10^{-18}$ C	Planck charge
$T_p = 1.41678698590795 \times 10^{32}$ K	Planck temperature
$m_p = 2.17642683817579 \times 10^{-8}$ kg	Planck mass
$\Xi = 0.999999199973626 \times 10^{-7}$	hyperbolic inversion boundary
5	number of dimensions
7	break in scale symmetry
$\pi_1 = 0.0854254531533304 \dots$	1 st hyperbolic partition constant
$\pi_2 = 3.66756753485501 \dots$	2 nd hyperbolic partition constant
$\pi_3 = 1.87649603900417 \dots + 0.46615262615972 \dots i$	3 rd hpc
$\pi_4 = 1.87764603900417 \dots - 0.46615262615972 \dots i$	4 th hpc
$\pi_5 = 4.48762644916751 \dots$	hyperbolic radius constant
$\pi_6 = 2.00316562310924 \dots$	hyperbolic radian constant
$G_{00} = 1.01494160640965 \dots$	Gieseking's constant
$V_6 = 0.209598321821930 \dots$	figure eight knot hyperbolic volume
$K = 0.9159655841772190 \dots$	Catalan's constant
$C_d = 0.2916506904030818 \dots$	domino tiling constant
$C_4 = 1.79162281206959 \dots$	dimer constant
$\omega_1 = 0.7649747018528596 \dots + 1.3249706214087 \dots i$	omega_1 constant
$\omega_2 = 1.529594037057192 \dots$	omega_2 constant
$i^{i^{i^i}} = 0.438282936727032 \dots + 0.360592471871385 \dots i$	power tower of i
$i = (-1)^{1/2}$	imaginary unit
$\phi_1 = (-1)^{1/3}$	imaginary golden ratio
$G_9 = 1.15872847301812 \dots$	tether length for grazing half the unit circle
$\pi = 3.14159265358979 \dots$	Archimedes' constant
$S = 2.29558714939263 \dots$	universal parabolic constant
$s = 5.244115110858423 \dots$	arc length of the unit lemniscate
$L = 2.620257554292119 \dots$	lemniscate constant
$L_1 = 1.31102877414605 \dots$	1 st lemniscate constant
$L_2 = 0.5949701117367796 \dots$	2 nd lemniscate constant
$C_{ij} = 0.889213084793979 \dots$	ubiquitous constant
$G_{06} = 0.834626841674073 \dots$	Gauss's constant
$C_{CF} = 0.697777456496189 \dots$	continued fraction constant
$R_{FR} = 0.99813604459805 \dots$	Ramanujan's 1 st continued fraction constant
$F_{FF} = 1.22674201072035 \dots$	Fibonacci factorial constant
$M = -1.74756459463318 \dots$	Modeling constant
$e = 2.71828182845904 \dots$	Euler's number
$\gamma = 0.577215664901532 \dots$	Euler-Mascheroni constant
$S = 0.822862549678847 \dots$	Sierpinski's constant
$C_{JP} = 2.4048825650575772 \dots$	1 st root of the Bessel function
$J_{011} = 1.19967864025773 \dots$	Real fixed point of the hyperbolic cotangent
$D_{N0} = 0.739085133216160 \dots$	Dottie number
$L = 0.6627431934931581 \dots$	Laplace limit
$\alpha_F = 2.50290787590589 \dots$	alpha Feigenbaum constant
$\delta_F = 4.66920160910299 \dots$	delta Feigenbaum constant
$S = 3.246497960371714 \dots$	silver constant
$P = 1.32471795724474 \dots$	plastic constant
$Q = 0.605443657196732 \dots$	QRS constant
$C_{QA} = 2.03816397970215 \dots$	associated QRS constant
$\chi_{\text{ISM}} = 0.353236371854995 \dots$	Hafner-Sarnak-McCurley constant
$T_0 = 2.731500000000000 \times 10^2$ kelvin	freezing point of water
$P_0 = 1.000000000000000 \times 10^5$ pascal	1 bar
$P_1 = 1.01325003754773 \times 10^5$ pascal	standard atmospheric pressure
$E_1 = 6.09110229711386 \times 10^{-24}$ joules	energy associated with $\Delta\chi_{\text{KS}}$
$f = 5.400000000000000 \times 10^{-14}$ Hz	frequency of max/light wavelength
$\lambda_{\text{peak}} = 4.96511423174427 \dots$	peak λ of spectral radiance/unit wavelength
$\nu_{\text{peak}} = 2.8214393712207 \dots$	peak emission of spectral flux/unit frequency
$\log(x) =$ natural logarithm	
$\Gamma(x) =$ gamma function	ln = lumen = cd · sr
$\zeta(x) =$ Riemann zeta function	$N =$ newton = m kg/s ²
$\Gamma =$ derangement function	$N =$ noether = m kg/s
	$\nu =$ ohm = m ² kg/s ² C ²
units	composite units
$m =$ meter	$A =$ ampere = C/s
$s =$ second	$F =$ farad = s ² C ² /m ² kg
$C =$ coulomb	$H =$ henry = m ² kg/C ²
$K =$ kelvin	$\text{Hz} =$ hertz = 1/s
$\text{kg} =$ kilogram	$J =$ joule = m ² kg/s ²
derived units	
MHz = megahertz = 1.000000000000000 $\times 10^6$ 1/s	
fm = femtometer = 1.000000000000000 $\times 10^{-15}$ m	
eV = electron-volt = 1.000000000000000 eV	
MeV = megaelectron-volt = 1.000000000000000 $\times 10^6$ eV	
GeV = gigaelectron-volt = 1.000000000000000 $\times 10^9$ eV	
$\frac{r_n}{r_e} = \left(\frac{4\pi}{rS}\right)^2$	$\frac{r_n}{r_e} = \frac{C_0}{4}$
$\frac{r_n}{r_e} = \frac{r_n}{r_e}$	$\frac{r_e}{r_0} = \sqrt{\frac{2\pi}{\log(36)}}$
$\epsilon_p = \frac{l^2 m_p}{\epsilon_r r^2}$	
external transform space	Physics Monastery
$V_{2\pi} = \left(\frac{4\pi}{5t}\right) \left(\frac{4\pi}{15}\right) \left(\frac{4\pi}{35}\right) \left(\frac{4\pi}{18}\right) \left(\frac{4\pi}{32}\right) \left(\frac{4\pi}{8}\right)^4$	Thad Roberts
figure eight knot hyperbolic volume	October-26-2025
$V_0 = \left\{ \left(L_2 \left(\frac{1}{\phi_1} \right) - L_2 \left(\phi_1 \right) \right) \right\}$	with: